

Coringtreepotters ms

Christophe Rouleau-Desrochers

July 3, 2026

Methods

Citizen science program

The Tree Spotters was a citizen science program that started in 2015 and aimed to train citizen scientists for accurate and rigorous phenological monitoring at the Arnold Arboretum of Harvard University (42.30 °N, -71.12 °W; Figure 1). From 2015 to 2024, hundreds of citizen scientists monitored 77 individual trees and shrubs of 14 species. They regularly followed those individuals from budburst in the spring to leaf coloring in the fall using the National Phenology Network (NPN) phenophases (Denny *et al.*, 2014): Leaves (483), Colored leaves (498), Fruits (516), Ripe Fruits (390), Falling leaves (471), Recent fruit or seed drop (504), Increasing leaf size (467), Breaking leaf buds (371), Flowers or flower buds (500), Open flowers (501), Pollen release (502). Not all phenophases were recorded for every tree, for every year, and some trees miss several years of data. However, here we use leafout to colored leaves (or leaf coloring) to define the growing season length. Taken together, our dataset comprises 321 leafout and 284 colored leaves observations (which we refer to as the ‘full datasets’), but we did not have both observations for all trees in all years, so our total of matched full growing seasons duration (which we refer to as the ‘restricted dataset’) is 278. We visually checked for SOS, EOS and GSL outliers and verified that there was no observation bias potentially caused by a particular observer or in a particular year.

Phenological monitoring and sample collection

From 20 to 22 April 2025, we collected tree ring samples only for the tree species that were previously monitored for phenology—excluding *Fagus grandifolia* because of beach bark disease concerns—for a total of 49 individuals (Table S1). Thus, we sampled four or five individuals for 11 species: Red maple (*Acer rubrum*), Sugar maple (*Acer saccharum*), Yellow buckeye (*Aesculus flava*), Yellow birch (*Betula alleghaniensis*), River birch (*Betula nigra*), Pignut hickory (*Carya glabra*), Shagbark hickory (*Carya ovata*), Eastern cottonwood (*Populus deltoides*), White oak (*Quercus alba*), Northern red oak (*Quercus rubra*) and American basswood (*Tilia americana*). For this, we collected two 5-mm diameter cores, 15-cm length at 1.3 meters above ground, perpendicular to the slope and at 180 degrees from each other, using an increment borer (Mora Borer, Haglöfs Sweden, Bromma, Stockholm, Sweden). We cleaned the increment borer with alcohol (70% ethanol) and the inside with a brush before collecting each core. We stored the cores in modified plastic straws to allow aeration and were left to dry at ambient temperature for three months.

Sample processing, imaging and measuring

We used wooden mounts to stabilize the cores that were then sanded using progressively finer sandpaper grit (from 150 to 1000) by increments of roughly 150. We then used a high-resolution scanner to scan the cores at a resolution of 6250 dpi (Fong *et al.*, 2026), and used those images to measure the ring widths using ImageJ (Schneider *et al.*, 2012).

We visually cross-dated the ring width chronologies by assigning a calendar year to each ring and comparing yearly variation of trends across the different individuals of the same species (Cook & Kairiukstis, 1990). Our short chronologies and low species replication prevented us from performing statistical cross-dating. We

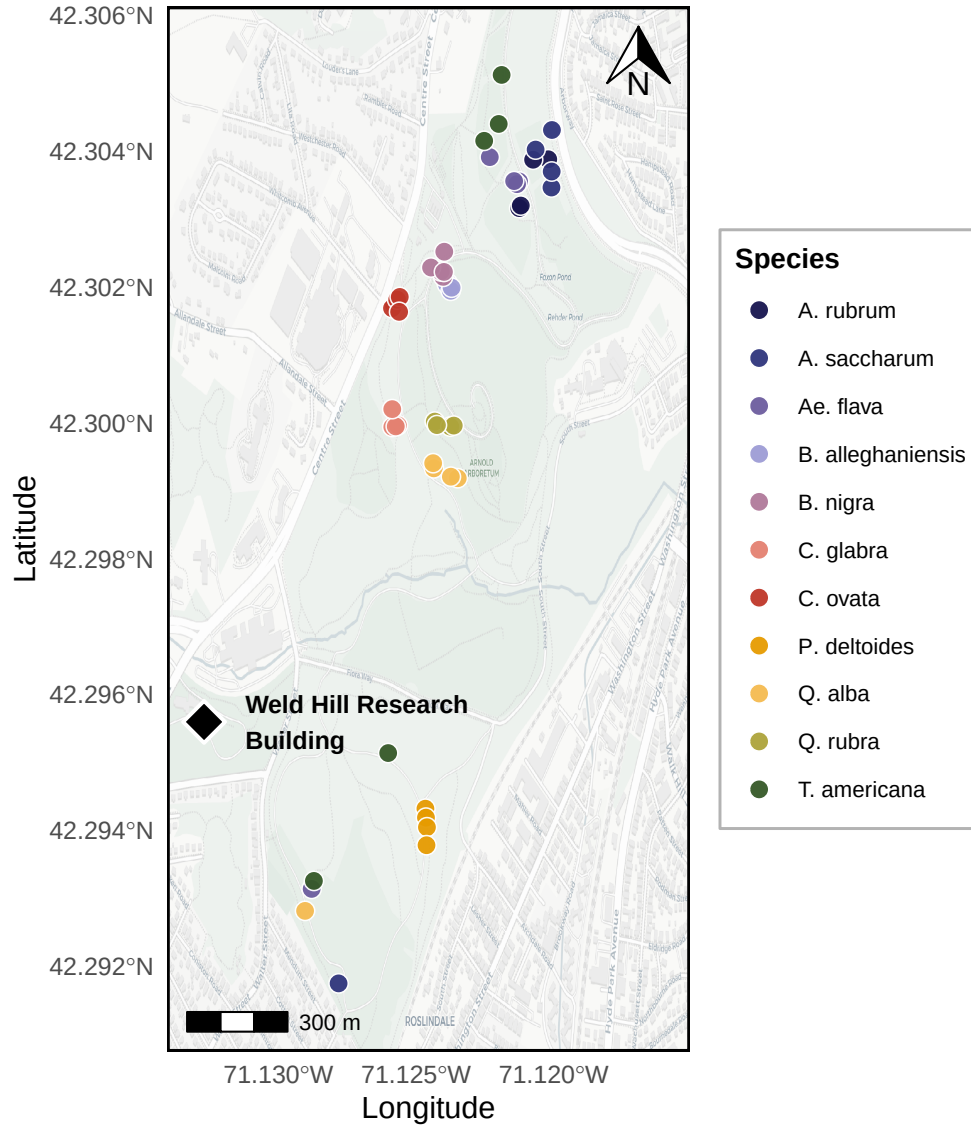


Figure 1: Map of the Arnold Arboretum of Harvard University with the locations of the trees depicted as the dots. The colors of the dots match the model estimates of model estimate figures (e.g. Figure 2). We also show the location of the Weld Hill Research Building (black diamond), where the common garden from Chapter 1 took place.

also did not detrend our chronologies because the ring widths during our study period (nine years) showed no consistent allometric trend (Figure S1). We included year in our models to control for effects of different years.

Data analyses

We used Bayesian hierarchical models to understand how growth shifted with four different season metrics: GDD, GSL, SOS and EOS. We used growing degree days (GDD; mean daily temperature above 5°C) between leafout and leaf coloring to calculate the thermal season and the number of days between leafout and leaf coloring for the calendar season (growing season length; GSL). Then, we used leafout for the start of

season (SOS) and leaf coloring for the end of season (EOS). To better illustrate ring width responses to our season metrics, we report ring width change ($\log(\text{mm})$) in response to the number of GDDs accumulated in an average week (seven days) of May (day of year 121 to 150 from 2016 to 2024; 58.32 GDD) for the thermal season. Similarly, we report ring width change per week of season length, leafout and leaf coloring for GSL, SOS and EOS, respectively. We report effects when the 90% UI do not overlap with zero across all models.

We combined daily climate data from two sources for the duration of our study (2016-2024). From 2016-2019, we used the Weld Hill research station climate station (Arnold Arboretum, 2026) and the "Boston (Weld Hill)" station for the remaining years (2020-2024; Carroll *et al.*, 2026). Both are located right next to the Arboretum. We determined drought occurrence at the arboretum using the North American Drought Monitor Maps and used the proposed drought index to qualify their intensity (National Centers for Environmental Information (NCEI), 2026).

Thus for each season predictor, we estimated its effect on ring width observed for each tree in each year as:

$$\begin{aligned} \ln(\text{ringwidth})_i &\sim \text{Normal}(\mu, \sigma_y) \\ \mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{\text{species}} \times \text{season predictor} \\ \alpha_{\text{treeid}} &\sim \text{Normal}(0, \sigma_{\alpha_{\text{treeid}}}) \end{aligned} \quad (1)$$

With this model, we estimated a species-specific effect of each season predictor on growth (β_{species}), but we also controlled for variation due to species (α_{species}), individual tree (α_{treeid}) and year (α_{year}). We first ran the models using natural units and then scaled the predictors through z -scoring so we could directly compare their effect size (Gelman & Hill, 2006). We report estimates as means and 90% uncertainty intervals (UI). We also fit the models for SOS and EOS using the full vs restricted datasets. Finally, because the SOS diverge across species with different wood anatomy, we ran the models using budburst instead of leafout as the SOS—allowing us to test whether our model responses diverged across diffuse vs ring porous species (Table S1).

We used an additional model to account for the previous year thermal conditions. For this, we used the GDD accumulated between leafout and leaf coloring from the previous year (e.g. for growth year of 2020, we used the GDD of 2019 as the previous year GDD), alongside the GDD accumulated during the current year. For interpretability, we also report ring width change ($\log(\text{mm})$) as the number of GDDs (for both years) accumulated in an average week (seven days) of May (day of year 121 to 150 from 2016 to 2024; 58.32 GDD). Thus, we estimated the previous and current year thermal season effect on the current year's growth as:

$$\begin{aligned} \ln(\text{ringwidth})_i &\sim \text{Normal}(\mu, \sigma) \\ \mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} \\ &\quad + \beta_{\text{species}}^{\text{current}} \times \text{GDD}_{\text{current}} + \beta_{\text{species}}^{\text{previous}} \times \text{GDD}_{\text{previous}} \\ \alpha_{\text{treeid}} &\sim \text{Normal}(0, \sigma_{\alpha_{\text{treeid}}}) \end{aligned} \quad (2)$$

While controlling for the same variation across species, individual tree and year, this model allowed us to account for the effect of the thermal season during the current year ($\beta_{\text{species}}^{\text{current}}$), but also during the previous year ($\beta_{\text{species}}^{\text{previous}}$).

Finally, we used a third Bayesian hierarchical model to determine whether the SOS and EOS of our species were related during the current year (we did not consider the effect of the EOS on the following year's SOS). For this, we estimated the effect of the SOS on the EOS on each tree, in each year as:

$$\begin{aligned} \text{EOS}_i &\sim \text{Normal}(\mu_i, \sigma_y) \\ \mu_i &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{\text{species}} \times \text{SOS} \\ \alpha_{\text{treeid}} &\sim \text{Normal}(0, \sigma_{\alpha_{\text{treeid}}}) \end{aligned} \quad (3)$$

Results

Our dataset comprised 11 deciduous tree species with leaf phenology data spanning nine years (2016 to 2024) for a total of 321 leafout and 284 leaf coloring observations, yielding 278 complete growing seasons (GS) for 49 individuals. Throughout the duration of the study, mean summer temperatures spanned 21.53°C (2017) to 23.07°C (2020), with an average of 22.51°C. 2020 and 2022 had moderate droughts in the summer (June, July and August), which ramped up to severe in September. Additionally, a moderate drought in June 2016, ramped up to severe in July, then extreme in August and September. The calendar season (GSL) ranged from 77 days (*A. flava* in 2018) to 194 days (*A. saccharum* in 2021), while the thermal season (GDD) ranged from 1048.8 GDD (*A. flava* in 2020) to 2634.7 GDD (*T. americana* in 2016). The earliest start of season (leafout) was 08-Apr (*A. flava* in 2021) and the latest, 15-May (*C. ovata* in 2020), while the earliest leaf coloring day of year was 15-Jul (*A. flava* in 2021—same individual with earliest leafout) and the latest, 06-Nov (*T. americana* in 2016). Annual ring widths across years varied from 0.43 mm (*A. saccharum* in 2019) to 11.75 mm (*T. americana* in 2019) (Figure S3).

Across our eleven species and season predictors, we did not find any clear response of the growing season length on growth. We report estimates as means and 90% uncertainty intervals (UI) that correspond to ring width change in mm on the log scale—henceforth ‘ Δ RW/scaled predictor’ (see ‘Data analyses’ section in Methods for more details on predictor scaling). Longer thermal season did not increase growth for any species and decreased growth for *B. alleghaniensis* ($\beta = -0.08$, UI: -0.12, -0.03 Δ RW/scaled GDD; Figure 2a and e; Table S2). Then, a longer calendar season did not change growth for any species (Figure 2b and f; Table S2), and on a scaled axis (z -scored), the thermal and calendar seasons were similarly predictive (Figure S11).

Previous year GDD increased growth for *B. alleghaniensis* ($\beta = 0.15$, UI: 0.02, 0.28 Δ RW/scaled GDD), and the current year GDD decreased its growth, like the observed response when the previous year conditions were not considered. *B. alleghaniensis*’s congeneric *B. nigra*, also grew less with previous year GDD ($\beta = -0.08$, UI: -0.17, -0.00 Δ RW/scaled GDD; Figure 3b; Table S5) but in contrast, current year GDD increased growth for that species ($\beta = 0.12$, UI: 0.03, 0.21 Δ RW/scaled GDD; Figure 3a; Table S5). Interestingly, *B. nigra* did not solely respond to current year’s conditions (Figure 2a and e; Table S2), but grew more with warmer thermal season when the previous conditions were considered.

An earlier start of season did not increase growth for any species and decreased growth for *T. americana* ($\beta = 0.15$, UI: 0.02, 0.27 Δ RW/scaled SOS; Figure 2c and g; Table S2), which did not respond to the thermal or the calendar season. Additionally, a later end of season did not increase growth for any species and decreased growth for *B. alleghaniensis* ($\beta = -0.09$, UI: -0.16, -0.01 Δ RW/scaled EOS; Figure 2d and h; Table S2), which also decreased growth with warmer thermal season. Then, the estimates were unchanged when using the full or restricted datasets (SOS: $n = 321$ and EOS: $n = 284$ vs $n = 278$; Figure S9b), or when using budburst instead of leafout as the start of the growing season (Figure S10c).

Finally, a later SOS delayed EOS only for *P. deltoides* ($\beta: 1.04$, UI: 0.39, 1.70 Δ EOS/SOS; Figure S12. We did not find any clear difference on baseline growth across years (Figure S2; Table S3) and between species (Figure S6; S7). Growth was similar across individual trees (Figure S8).

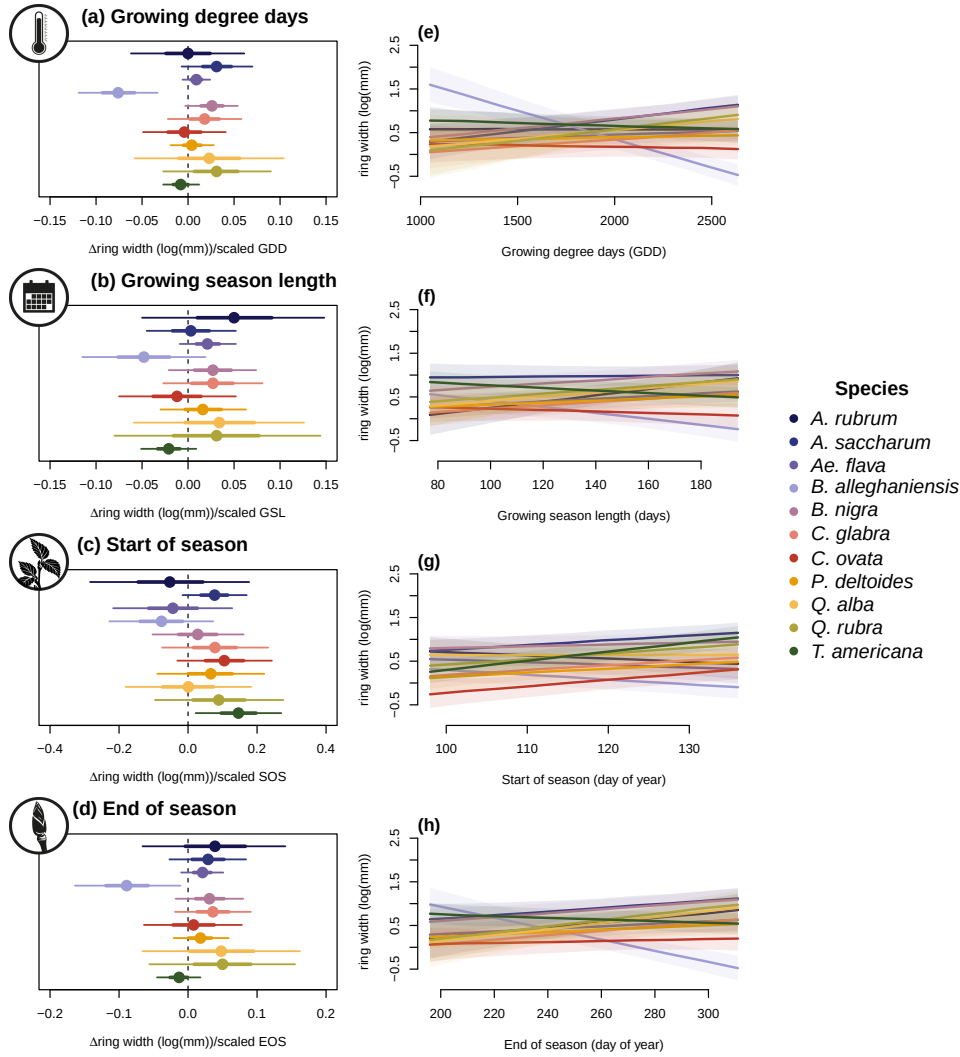


Figure 2: Estimates of ring width change (log(mm)) as a function of four different phenological predictors: growing degree days of the growing season (mean daily temperature above 5°C); growing season length; start of season; end of season across 11 species: (Red maple (*Acer rubrum*), Sugar maple (*Acer saccharum*), Yellow buckeye (*Aesculus flava*), Yellow birch (*Betula alleghaniensis*), River birch (*Betula nigra*), Pignut hickory (*Carya glabra*), Shagbark hickory (*Carya ovata*), Eastern cottonwood (*Populus deltoides*), White oak (*Quercus alba*), Northern red oak (*Quercus rubra*) and American basswood (*Tilia americana*)); species colors are the same across the panels). Panels (a) to (d) represent the model slope estimates for each species, for each predictor, which we represent through $\Delta \text{RW/scaled predictor}$ (see ‘Data analyses’ section in Methods for more details on predictor scaling). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI; see Table S2 for parameter estimates). Panels (e) to (h) represent the simulated response curves of ring width as a function of each predictor, with the lines depicting the mean response at each simulated predictor value, and the shaded area, the interquartile ranges (50% UI). These models do not include a slope on the previous year GDD (see Figure 3 for these estimates).

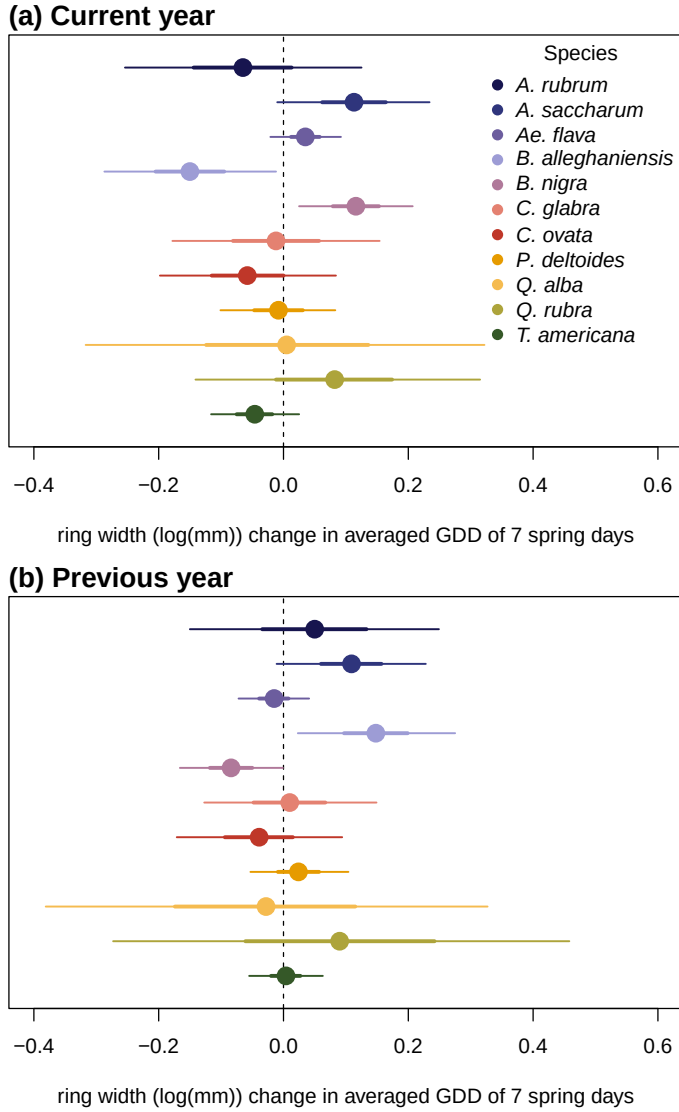


Figure 3: Model slope estimates representing ring width change (log(mm)) as a function of the current year growing degree days (GDD; a) and the previous year GDD (b). GDD represent the accumulated growing degree days (mean daily temperature above 5°C)) between leafout and leaf colouring and the models estimates shows ring width change (log(mm)) per scaled GDD (see ‘Data analyses’ section in Methods). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

Discussion

References

- Arnold Arboretum (2026). Weather Data at the Arnold Arboretum. Accessed: 2025-09-24.
- Burns, R.M., Honkala, B.H. & [Technical coordinators] (1990). *Silvics of North America: 2. Hardwoods*. United States Department of Agriculture (USDA), Forest Service, Agriculture Handbook 654.
- Carroll, J., Weigle, T. & Petzoldt, C. (2026). The Network for Environment and Weather Applications (NEWA). Accessed: 2025-09-24.
- Cook, E.R. & Kairiukstis, L.A. (eds.) (1990). *Methods of Dendrochronology*. Springer Netherlands, Dordrecht.
- Denny, E.G., Gerst, K.L., Miller-Rushing, A.J., Tierney, G.L., Crimmins, T.M., Enquist, C.A.F., Guertin, P., Rosemartin, A.H., Schwartz, M.D., Thomas, K.A. & Weltzin, J.F. (2014). Standardized phenology monitoring methods to track plant and animal activity for science and resource management applications. *International Journal of Biometeorology*, 58, 591–601.
- Fong, A., Curry, C., Zhang, Y. & Wolkovich, E. (2026). TIM: Tree Imaging Machine for high resolution rapid digital images of tree cores and cookies. *Unpublished*.
- Gelman, A. & Hill, J. (2006). *Data Analysis Using Regression and Multilevel/Hierarchical Models*. 1st edn. Cambridge University Press.
- National Centers for Environmental Information (NCEI) (2026). North American Drought Monitor. Short-author: NCEI Accessed: 2026-04-22.
- Schneider, C.A., Rasband, W.S. & Eliceiri, K.W. (2012). NIH Image to ImageJ: 25 years of image analysis. *Nature Methods*, 9, 671–675.

Supplemental

Supplemental Methods

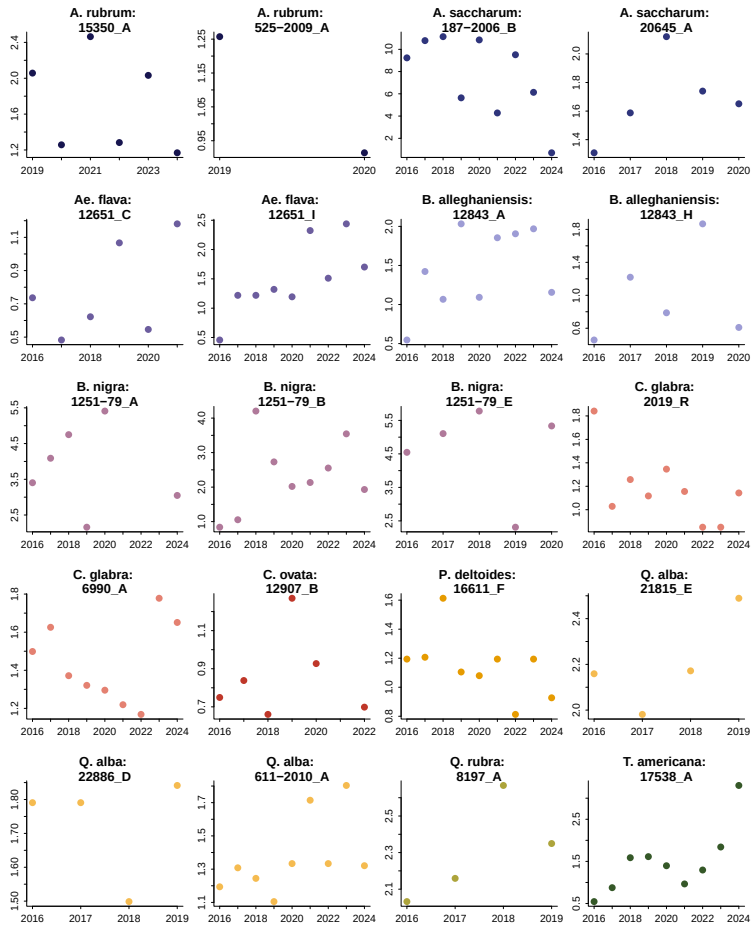


Figure S1: We randomly sampled 20 trees from our dataset and represent the ring width as a function of year to represent the allometry patterns across all years we had phenology data for.

Table S1: Treepotters species grouped by tree type, growth speed, and wood anatomy. We collected the growth speed information for each species from Burns *et al.* (1990) We sampled a total of 49 individual trees

Common Name (Latin binomial)	Growth speed	Wood Anatomy	<i>n</i>
American basswood (<i>Tilia americana</i>)	Fast	Diffuse-porous	5
Eastern cottonwood (<i>Populus deltoides</i>)	Fast	Diffuse-porous	4
Northern red oak (<i>Quercus rubra</i>)	Moderate	Ring-porous	4
White oak (<i>Quercus alba</i>)	Slow	Ring-porous	5
Pignut hickory (<i>Carya glabra</i>)	Slow	Ring-porous	4
Shagbark hickory (<i>Carya ovata</i>)	Moderate	Ring-porous	5
River birch (<i>Betula nigra</i>)	Fast	Diffuse-porous	4
Yellow birch (<i>Betula alleghaniensis</i>)	Slow	Diffuse-porous	4
Sugar maple (<i>Acer saccharum</i>)	Slow	Diffuse-porous	5
Red maple (<i>Acer rubrum</i>)	Moderate	Diffuse-porous	4
Yellow buckeye (<i>Aesculus flava</i>)	Moderate	Diffuse-porous	5

Supplemental results

Table S2: Species level slope parameters which represent ring width change ($\log(\text{mm})$), per scaled predictor (see ‘Data analyses’ section in Methods for more details on predictor scaling). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). Table S3 reports all the models parameters.

Species	GDD	GSL	SOS	EOS
<i>A. rubrum</i>	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
<i>A. saccharum</i>	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
<i>Ae. flava</i>	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
<i>B. alleghaniensis</i>	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
<i>B. nigra</i>	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
<i>C. glabra</i>	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
<i>C. ovata</i>	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
<i>P. deltoides</i>	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
<i>Q. alba</i>	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
<i>Q. rubra</i>	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
<i>T. americana</i>	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]

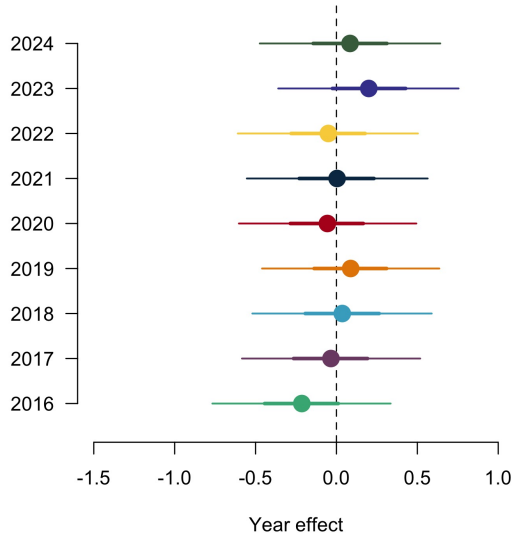


Figure S2: Ring width ($\log(\text{mm})$) deviations from the grand mean for each year using the GDD model (Table S3). The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

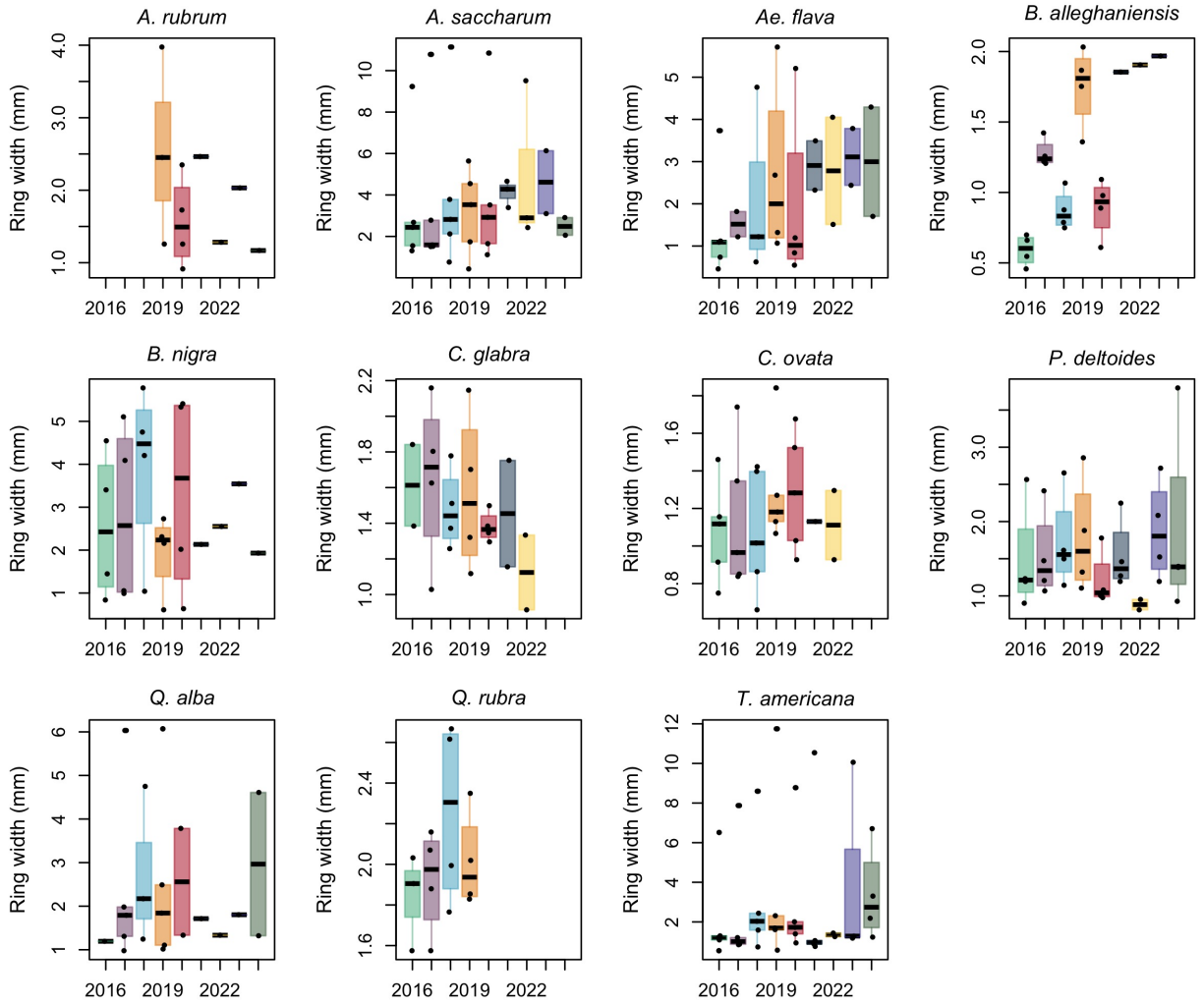


Figure S3: Boxplot representing the empirical data of our ring widths (mm) across years, faceted by species. The dots represent the raw data, the line the mean and the box the interquartile range (IQR).

Table S3: Summary output of each parameter from our four predictors, using ring width ($\log(\text{mm})$) as the response variable (See equation 1). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See ‘Data analyses’ section in Methods for more details.

Parameter	GDD	GSL	SOS	EOS
α	0.61 [-2.08, 3.25]	0.49 [-2.09, 3.03]	0.24 [-2.39, 2.94]	0.23 [-2.48, 2.98]
$\sigma_{\alpha_{tree}}$	0.55 [0.45, 0.67]	0.55 [0.45, 0.68]	0.54 [0.44, 0.66]	0.55 [0.45, 0.67]
σ_y	0.31 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]
$\beta_{speciesA.rubrum}$	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
$\beta_{speciesA.saccharum}$	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
$\beta_{speciesAe.flava}$	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
$\beta_{speciesB.alleghaniensis}$	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
$\beta_{speciesB.nigra}$	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
$\beta_{speciesC.glabra}$	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
$\beta_{speciesC.ovata}$	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
$\beta_{speciesP.deltoides}$	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
$\beta_{speciesQ.alba}$	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
$\beta_{speciesQ.rubra}$	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
$\beta_{speciesT.americana}$	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]
$\alpha_{year_{2016}}$	-0.05 [-3.40, 3.35]	-0.95 [-4.12, 2.17]	1.23 [-3.23, 5.60]	-1.08 [-5.62, 3.44]
$\alpha_{year_{2017}}$	-0.90 [-3.86, 2.10]	0.43 [-2.34, 3.17]	-0.57 [-3.59, 2.35]	-0.37 [-3.79, 3.02]
$\alpha_{year_{2018}}$	-0.49 [-3.20, 2.19]	-0.46 [-3.09, 2.14]	0.93 [-2.69, 4.47]	-0.49 [-3.33, 2.38]
$\alpha_{year_{2019}}$	2.32 [-0.68, 5.34]	0.61 [-2.29, 3.46]	1.17 [-2.34, 4.58]	3.25 [-0.47, 6.96]
$\alpha_{year_{2020}}$	-0.68 [-3.54, 2.12]	-0.13 [-2.82, 2.58]	0.17 [-3.18, 3.40]	-0.50 [-3.65, 2.67]
$\alpha_{year_{2021}}$	-0.89 [-3.89, 2.06]	-0.64 [-3.44, 2.12]	-1.17 [-4.65, 2.33]	-1.16 [-4.46, 2.12]
$\alpha_{year_{2022}}$	-0.32 [-3.35, 2.74]	-0.07 [-2.88, 2.73]	-1.96 [-5.42, 1.40]	-0.35 [-4.01, 3.29]
$\alpha_{year_{2023}}$	-0.38 [-3.14, 2.40]	-0.39 [-3.09, 2.26]	-1.04 [-4.60, 2.52]	-0.49 [-3.54, 2.50]
$\alpha_{year_{2024}}$	-0.85 [-4.70, 3.03]	-0.56 [-3.66, 2.52]	0.40 [-3.47, 4.32]	-1.44 [-6.32, 3.48]
$\alpha_{speciesA.rubrum}$	-1.09 [-4.47, 2.23]	-0.43 [-3.80, 2.96]	-1.08 [-4.98, 2.91]	-1.44 [-6.04, 3.22]
$\alpha_{speciesA.saccharum}$	0.30 [-2.41, 3.02]	0.59 [-1.98, 3.18]	-2.02 [-5.24, 1.14]	0.92 [-1.93, 3.81]
$\alpha_{speciesAe.flava}$	-0.21 [-0.77, 0.34]	-0.22 [-0.76, 0.35]	-0.25 [-0.80, 0.31]	-0.25 [-0.81, 0.29]
$\alpha_{speciesB.alleghaniensis}$	-0.03 [-0.58, 0.52]	-0.06 [-0.59, 0.52]	0.00 [-0.54, 0.56]	-0.07 [-0.63, 0.48]
$\alpha_{speciesB.nigra}$	0.04 [-0.52, 0.59]	0.02 [-0.52, 0.59]	-0.00 [-0.55, 0.56]	0.01 [-0.54, 0.56]
$\alpha_{speciesC.glabra}$	0.09 [-0.46, 0.64]	0.06 [-0.48, 0.63]	0.07 [-0.47, 0.64]	0.06 [-0.50, 0.62]
$\alpha_{speciesC.ovata}$	-0.06 [-0.60, 0.49]	-0.07 [-0.61, 0.50]	-0.12 [-0.66, 0.45]	-0.07 [-0.63, 0.47]
$\alpha_{speciesP.deltoides}$	0.00 [-0.55, 0.56]	-0.00 [-0.55, 0.57]	0.01 [-0.54, 0.58]	-0.01 [-0.58, 0.54]
$\alpha_{speciesQ.alba}$	-0.05 [-0.61, 0.50]	-0.07 [-0.63, 0.50]	-0.07 [-0.63, 0.49]	-0.08 [-0.65, 0.48]
$\alpha_{speciesQ.rubra}$	0.20 [-0.36, 0.76]	0.18 [-0.38, 0.75]	0.20 [-0.35, 0.78]	0.17 [-0.40, 0.72]
$\alpha_{speciesT.americana}$	0.08 [-0.47, 0.64]	0.07 [-0.47, 0.66]	0.07 [-0.48, 0.64]	0.08 [-0.49, 0.64]

Table S4: Summary output of each parameter from the previous year model (Equation 2) using ring width (log(mm)) as the response variable. We report estimates as mean and 90% uncertainty intervals (in square brackets). see ‘Data analyses’ section in Methods for more details

Parameter	Estimate
α	0.20 [-1.10, 1.48]
$\sigma_{\alpha_{tree}}$	0.58 [0.47, 0.72]
σ_y	0.29 [0.26, 0.32]
$\beta_{current}^{species A. rubrum}$	-0.07 [-0.25, 0.12]
$\beta_{current}^{species A. saccharum}$	0.11 [-0.01, 0.23]
$\beta_{current}^{species Ae. flava}$	0.04 [-0.02, 0.09]
$\beta_{current}^{species B. alleghaniensis}$	-0.15 [-0.29, -0.01]
$\beta_{current}^{species B. nigra}$	0.12 [0.03, 0.21]
$\beta_{current}^{species C. glabra}$	-0.01 [-0.18, 0.15]
$\beta_{current}^{species C. ovata}$	-0.06 [-0.20, 0.08]
$\beta_{current}^{species P. deltoides}$	-0.01 [-0.10, 0.08]
$\beta_{current}^{species Q. alba}$	0.01 [-0.32, 0.32]
$\beta_{current}^{species Q. rubra}$	0.08 [-0.14, 0.32]
$\beta_{current}^{species T. americana}$	-0.05 [-0.12, 0.03]
$\beta_{previous}^{species A. rubrum}$	0.05 [-0.15, 0.25]
$\beta_{previous}^{species A. saccharum}$	0.11 [-0.01, 0.23]
$\beta_{previous}^{species Ae. flava}$	-0.01 [-0.07, 0.04]
$\beta_{previous}^{species B. alleghaniensis}$	0.15 [0.02, 0.28]
$\beta_{previous}^{species B. nigra}$	-0.08 [-0.17, -0.00]
$\beta_{previous}^{species C. glabra}$	0.01 [-0.13, 0.15]
$\beta_{previous}^{species C. ovata}$	-0.04 [-0.17, 0.09]
$\beta_{previous}^{species P. deltoides}$	0.02 [-0.05, 0.10]
$\beta_{previous}^{species Q. alba}$	-0.03 [-0.38, 0.33]
$\beta_{previous}^{species Q. rubra}$	0.09 [-0.27, 0.46]
$\beta_{previous}^{species T. americana}$	0.00 [-0.06, 0.06]
$\alpha_{year_{2016}}$	0.44 [-3.10, 3.90]
$\alpha_{year_{2017}}$	-2.20 [-4.79, 0.36]
$\alpha_{year_{2018}}$	0.04 [-1.53, 1.59]
$\alpha_{year_{2019}}$	-0.06 [-2.61, 2.44]
$\alpha_{year_{2020}}$	0.30 [-1.39, 1.99]
$\alpha_{year_{2021}}$	0.18 [-1.77, 2.13]
$\alpha_{year_{2022}}$	1.16 [-1.18, 3.47]
$\alpha_{year_{2023}}$	-0.01 [-1.92, 1.90]
$\alpha_{year_{2024}}$	0.97 [-5.55, 7.68]
$\alpha_{species A. rubrum}$	-1.73 [-8.12, 4.60]
$\alpha_{species A. saccharum}$	0.93 [-0.65, 2.50]
$\alpha_{species Ae. flava}$	0.07 [-0.48, 0.62]
$\alpha_{species B. alleghaniensis}$	0.06 [-0.50, 0.61]
$\alpha_{species B. nigra}$	-0.30 [-1.02, 0.42]
$\alpha_{species C. glabra}$	-0.04 [-0.58, 0.53]
$\alpha_{species C. ovata}$	0.00 [-0.55, 0.56]
$\alpha_{species P. deltoides}$	-0.00 [-0.56, 0.56]
$\alpha_{species Q. alba}$	0.17 [-0.40, 0.73]
$\alpha_{species Q. rubra}$	0.01 [-0.56, 0.58]
$\alpha_{species T. americana}$	-0.05 [-0.60, 0.51]

Table S5: Summary output of each parameter from the model on the effect of SOS (days) on EOS (days; Equation 3). We report estimates as mean and 90% uncertainty intervals (in square brackets). see ‘Data analyses’ section in Methods for more details

Parameter	Estimate
α	29.51 [23.54, 35.39]
$\sigma_{\alpha_{treeid}}$	1.18 [0.83, 1.56]
σ_y	2.06 [1.90, 2.23]
$\beta_{species_{A.rubrum}}$	0.63 [-0.14, 1.42]
$\beta_{species_{A.saccharum}}$	0.18 [-0.29, 0.66]
$\beta_{species_{Ae.flava}}$	0.38 [-0.32, 1.08]
$\beta_{species_{B.allegghaniensis}}$	0.50 [-0.15, 1.16]
$\beta_{species_{B.nigra}}$	0.44 [-0.16, 1.06]
$\beta_{species_{C.glabra}}$	0.48 [-0.17, 1.15]
$\beta_{species_{C.ovata}}$	0.40 [-0.22, 1.01]
$\beta_{species_{P.deltoides}}$	1.04 [0.39, 1.70]
$\beta_{species_{Q.alba}}$	0.35 [-0.34, 1.04]
$\beta_{species_{Q.rubra}}$	0.50 [-0.19, 1.22]
$\beta_{species_{T.americana}}$	0.46 [-0.14, 1.05]
$\alpha_{year_{2016}}$	-2.26 [-15.00, 10.18]
$\alpha_{year_{2017}}$	7.21 [-1.83, 16.23]
$\alpha_{year_{2018}}$	0.36 [-10.75, 11.25]
$\alpha_{year_{2019}}$	-0.23 [-11.38, 10.91]
$\alpha_{year_{2020}}$	0.27 [-10.43, 10.84]
$\alpha_{year_{2021}}$	-0.20 [-11.26, 10.69]
$\alpha_{year_{2022}}$	1.92 [-9.19, 13.03]
$\alpha_{year_{2023}}$	-11.53 [-23.17, 0.25]
$\alpha_{year_{2024}}$	3.61 [-8.48, 15.48]
$\alpha_{species_{A.rubrum}}$	0.14 [-12.16, 12.19]
$\alpha_{species_{A.saccharum}}$	-0.27 [-10.42, 9.76]
$\alpha_{species_{Ae.flava}}$	1.86 [-0.38, 4.14]
$\alpha_{species_{B.allegghaniensis}}$	1.17 [-1.06, 3.41]
$\alpha_{species_{B.nigra}}$	0.47 [-1.80, 2.72]
$\alpha_{species_{C.glabra}}$	-0.42 [-2.70, 1.80]
$\alpha_{species_{C.ovata}}$	-1.02 [-3.27, 1.27]
$\alpha_{species_{P.deltoides}}$	-0.23 [-2.54, 2.06]
$\alpha_{species_{Q.alba}}$	-0.87 [-3.21, 1.45]
$\alpha_{species_{Q.rubra}}$	-0.28 [-2.63, 2.06]
$\alpha_{species_{T.americana}}$	-1.50 [-3.85, 0.81]

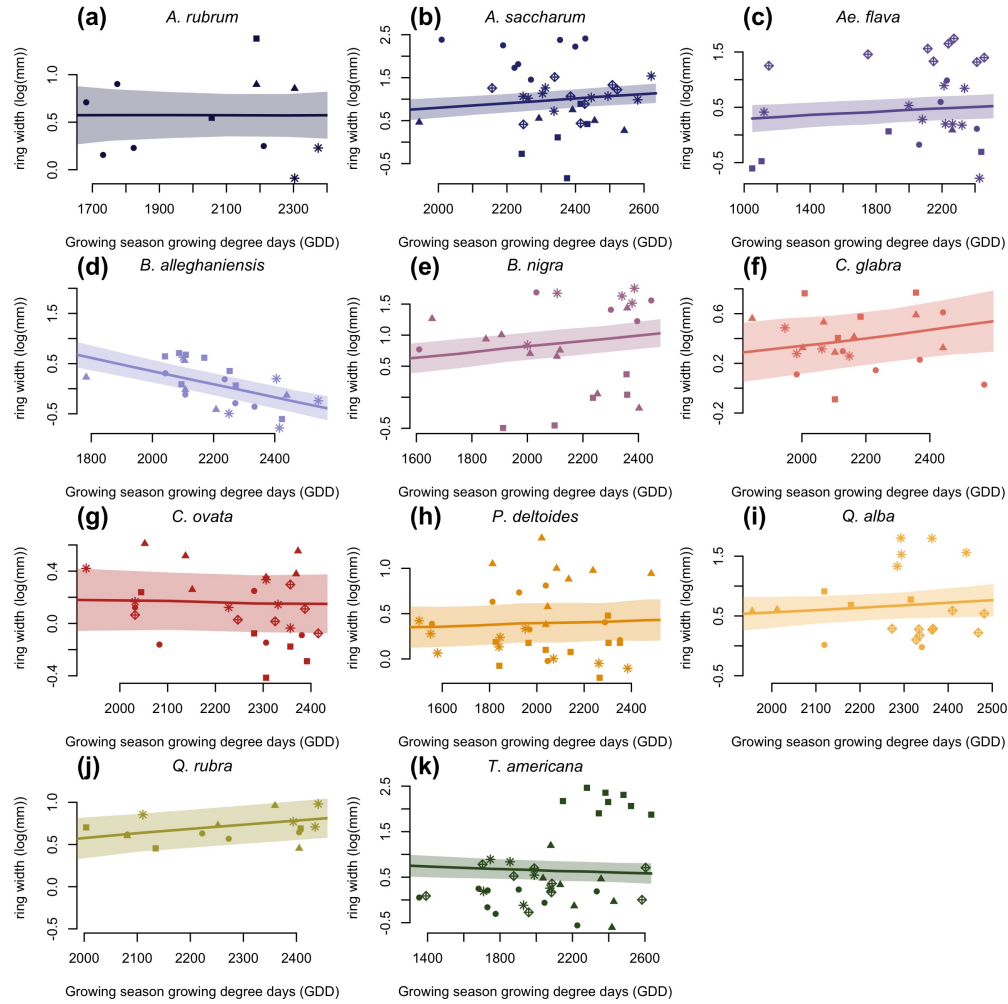


Figure S4: Ring width (log(mm)) trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

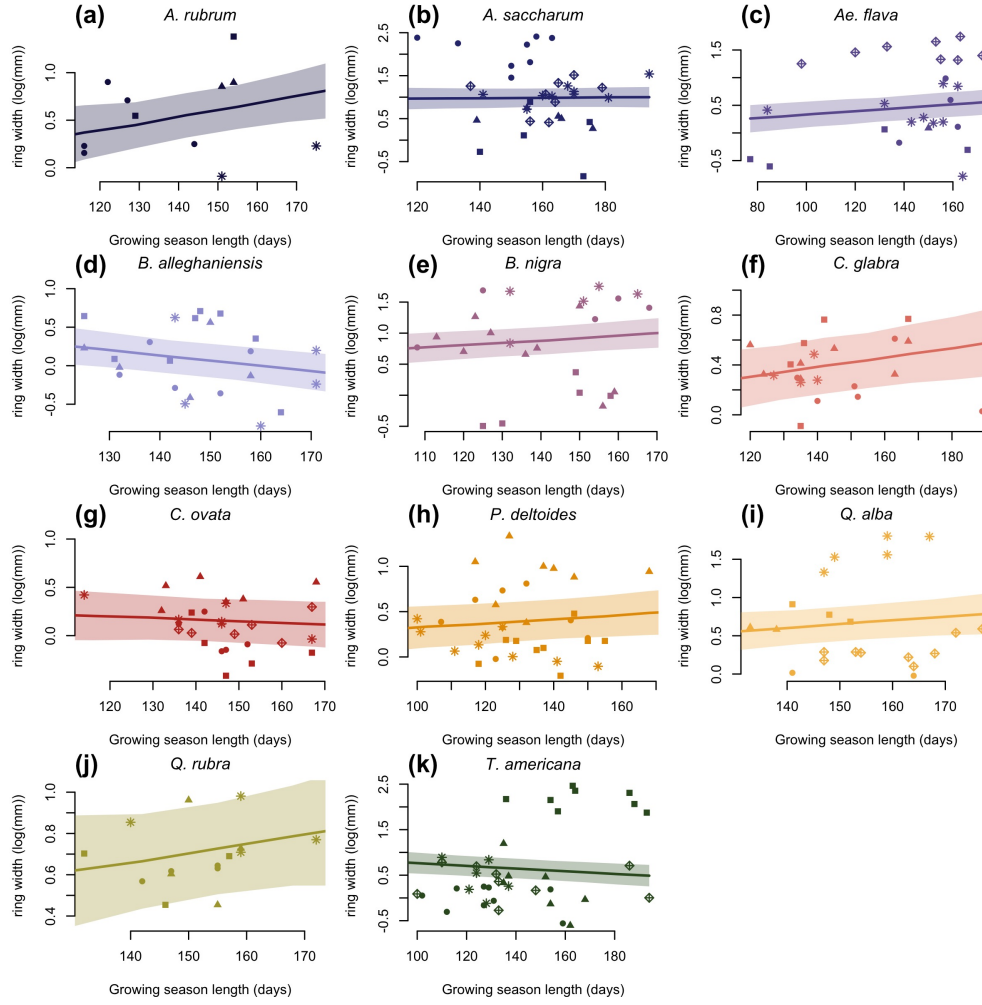


Figure S5: Ring width trends (log(mm)) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

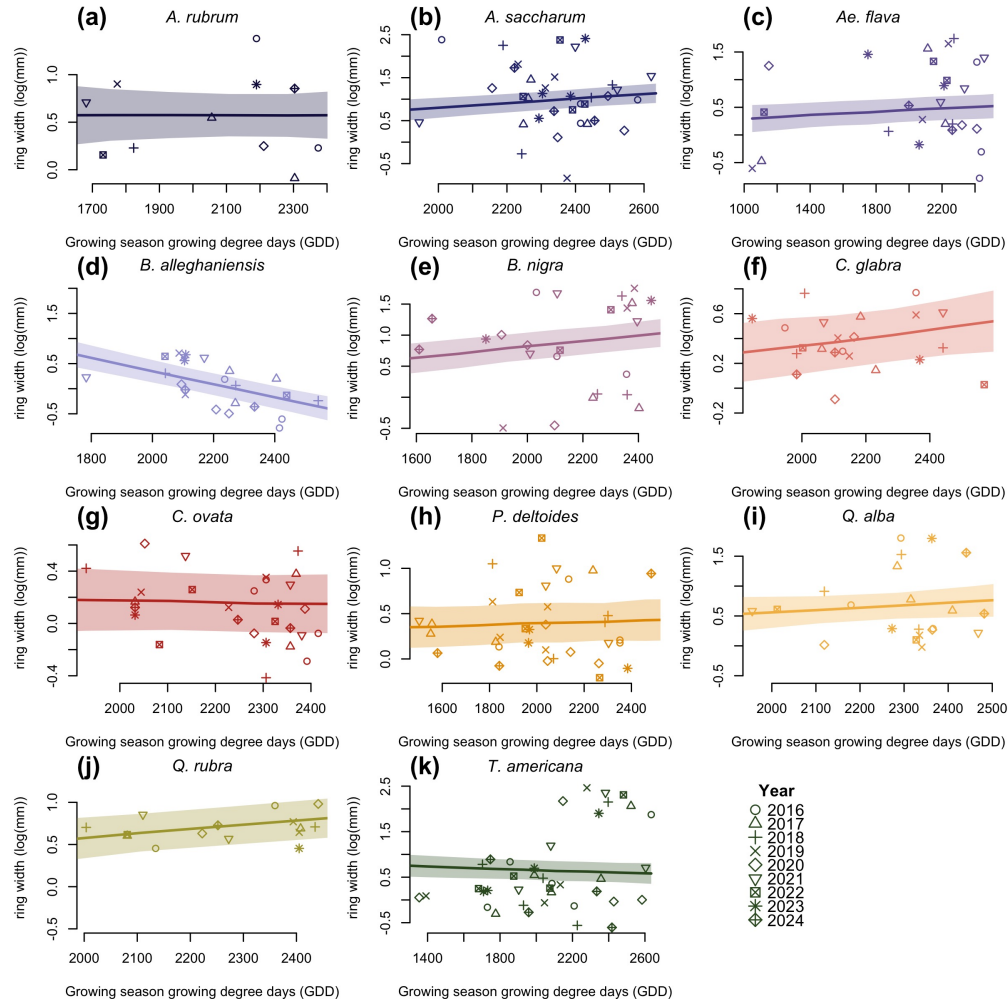


Figure S6: Ring width trends (log(mm)) in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

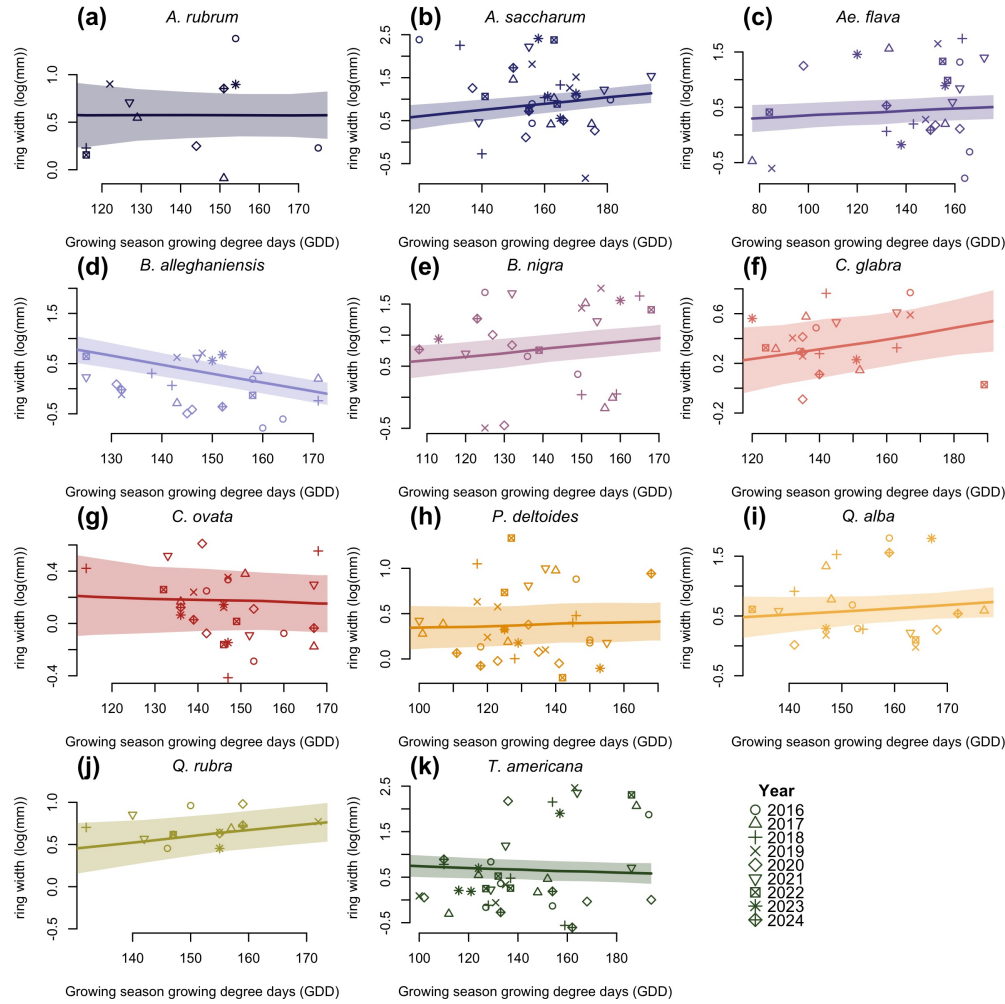


Figure S7: Ring width trends (log(mm)) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

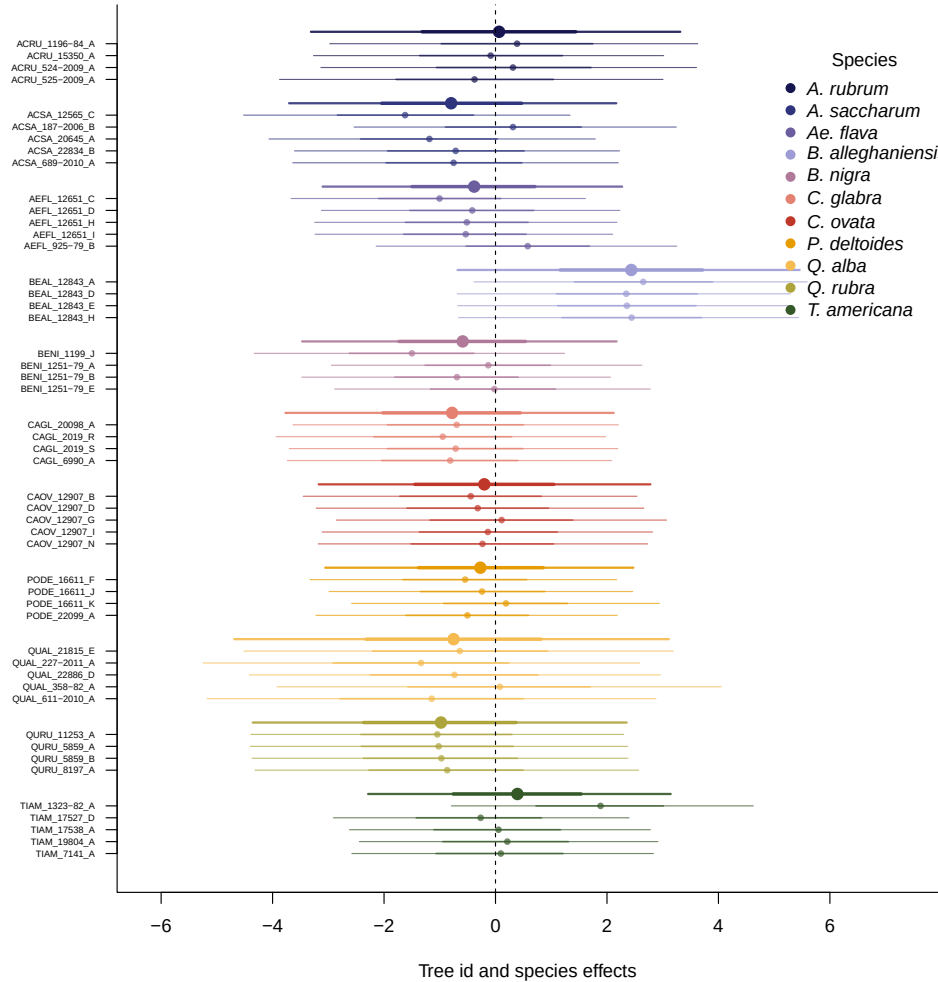
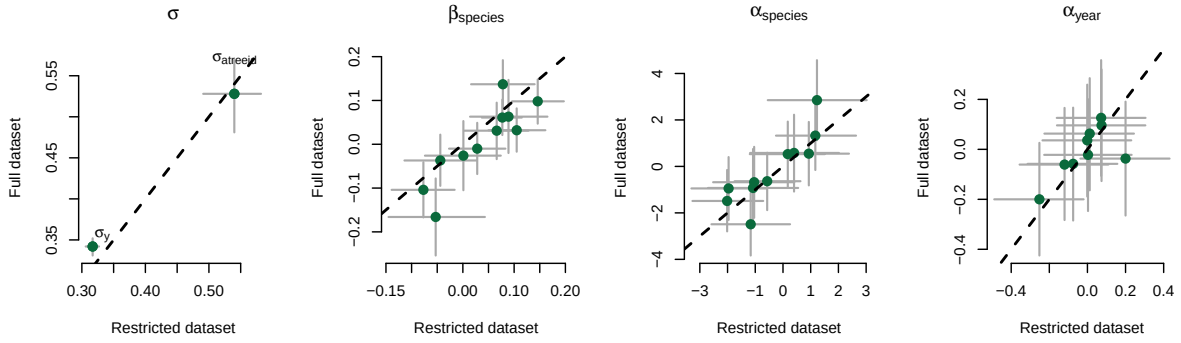


Figure S8: Ring width (log(mm)) deviations from the grand mean for each grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

(a) Start of season (SOS)



(b) End of season (EOS)

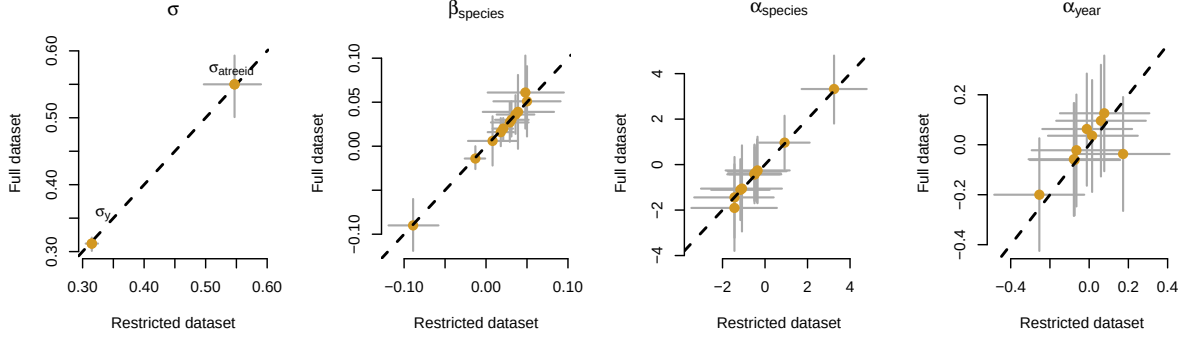


Figure S9: Full vs. most restricted rows of data for the model with start of season (SOS; a) and end of season (EOS; b) as the predictor. Each panel represent the different parameters used to fit the models. $\alpha_{species}$ and α_{year} represent the ring width (log(mm)) for species and year, respectively. $\beta_{species}$ represent the change in ring width (log(mm)) per scaled predictor for each species (see ‘Data analyses’ section in Methods for more details on predictor scaling).

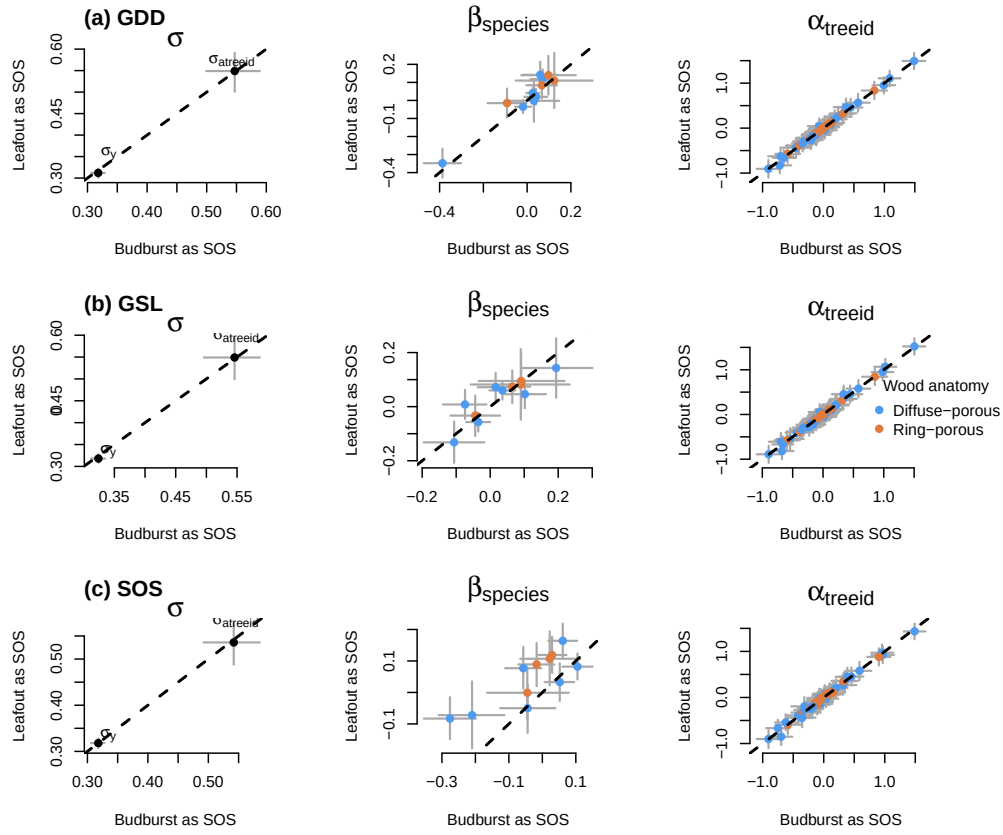


Figure S10: Parameter estimates when calculating the GDD (a), GSL (b) from leafout vs budburst, to leaf coloring. (c) shows the model estimates when taking leafout vs budburst as the SOS (see ‘Data analyses’ section in Methods for more details).

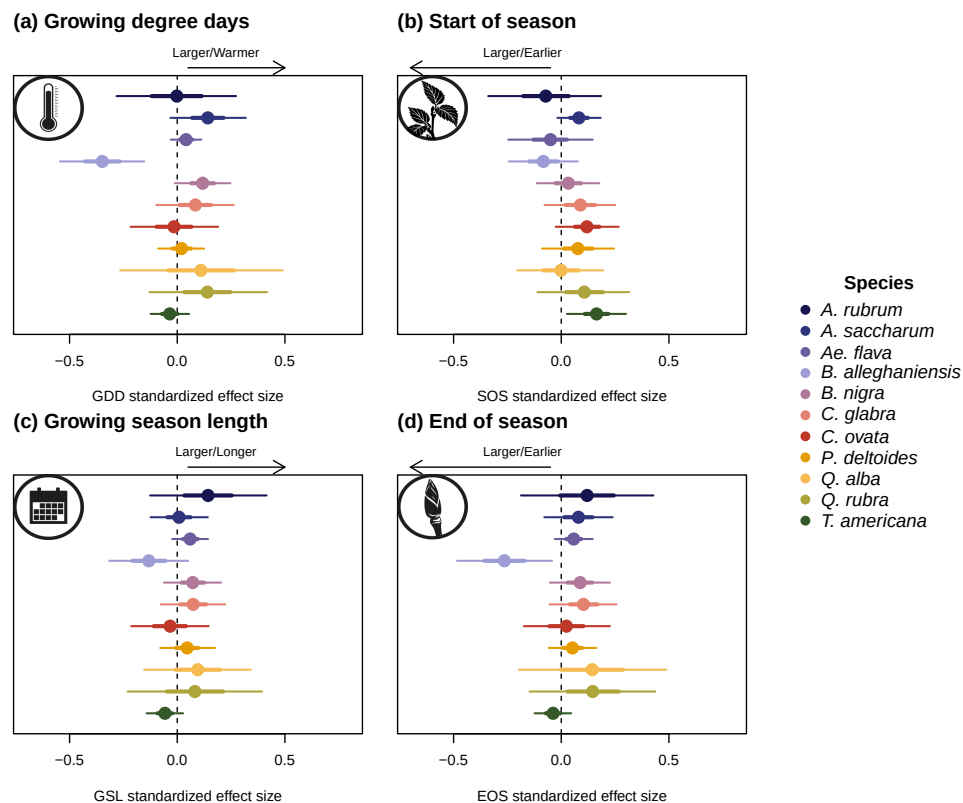


Figure S11: Effect size of each predictors across our eleven species. We depict ring width ($\log(\text{mm})$) change per scaled (z -scored; see ‘Data analyses’ section in Methods for more details) predictor for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

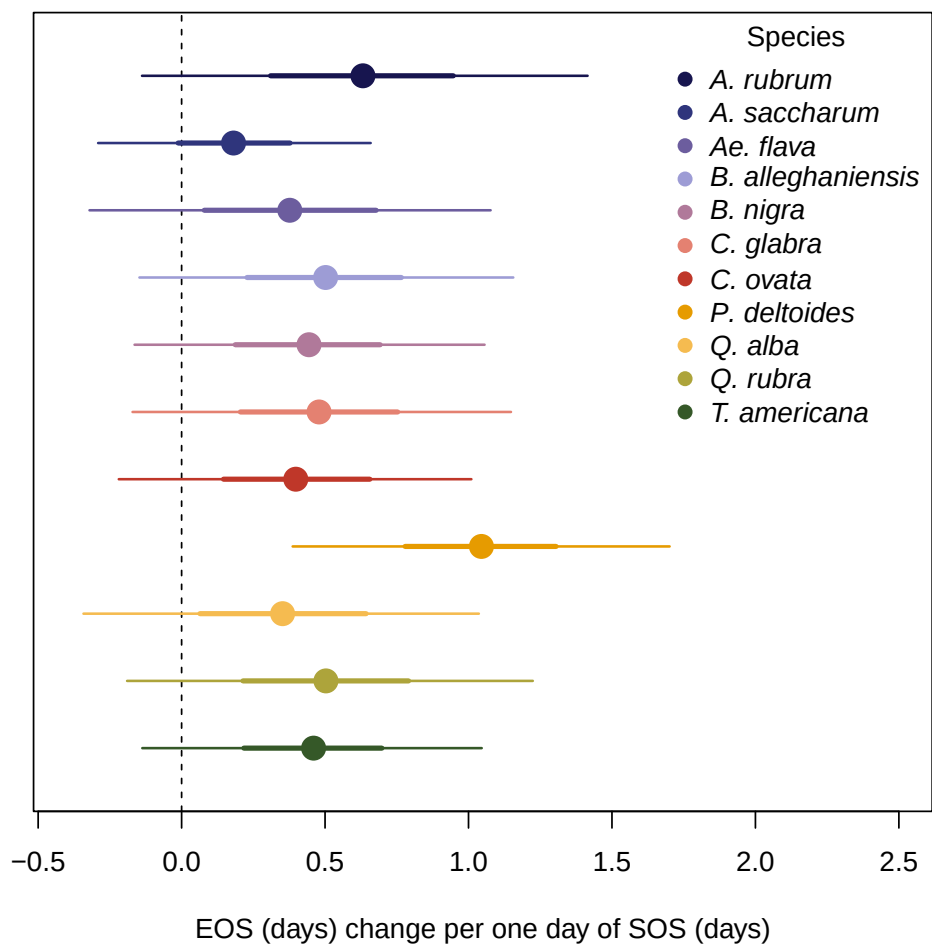


Figure S12: Effect of the start of season (SOS) on the end of season (EOS) across our eleven species. We report these slope estimates as mean and 90% uncertainty intervals.